

| Procedure                                                                                                                                                                                                    | Stages  | Cpl        | Loc       | Blk       | Op    | Collective<br>C sq S/W | Blocked resources<br>and remarks                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|------------|-----------|-----------|-------|------------------------|------------------------------------------------------------------------------------|
| MPI_BCAST and others                                                                                                                                                                                         | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                | 4) 18)                                                                             |
| MPI_IBCAST and others                                                                                                                                                                                        | i-s---  | ic         | l         | nb        | nb-op | C sq                   | buffer                                                                             |
| MPI_IGATHERV and other ... <b>V</b> / ... <b>W</b>                                                                                                                                                           | i-s---  | ic         | l         | nb        | nb-op | C sq                   | buffer, array arguments                                                            |
| corresponding MPI_WAIT                                                                                                                                                                                       | ---c-f  | c+f        | nl        |           | nb-op | C W2                   | 4) 5) 18)                                                                          |
| corr. MPI_TEST returning flag=TRUE                                                                                                                                                                           | ---c-f  | c+f        | l         |           | nb-op | C W2                   | 4) 5) 18)                                                                          |
| corr. MPI_TEST returning flag=FALSE                                                                                                                                                                          | -----   |            | l         |           | nb-op |                        | buffer, array arguments 7)                                                         |
| MPI_BCAST_INIT and others                                                                                                                                                                                    | i-----  | <b>ic</b>  | <b>nl</b> | <b>b</b>  | p-op  | C sq W1                | buffer address <b>8</b> ) 9)                                                       |
| MPI_GATHERV_INIT and other ... <b>V</b> /... <b>W</b> _INIT                                                                                                                                                  | i-----  | <b>ic</b>  | <b>nl</b> | <b>b</b>  | p-op  | C sq W1                | buffer address,<br>array arguments <b>8</b> ) 9)                                   |
| corresponding MPI_START, MPI_STARTALL                                                                                                                                                                        | -s---   | ic         | l         | <b>nb</b> | p-op  | C                      | buf.addr.+content 7) <b>14</b> ) 16)                                               |
| corresponding MPI_WAIT                                                                                                                                                                                       | ---c-   | c          | nl        |           | p-op  | C W2                   | buffer address and array<br>arguments cached on the request<br>handle, 4,5,7,9,18) |
| corr. MPI_TEST returning flag=TRUE                                                                                                                                                                           | ---c-   | c          | l         |           | p-op  | C W2                   | buf-addr&arr-args 4,5,7,9,18)                                                      |
| corr. MPI_TEST returning flag=FALSE                                                                                                                                                                          | -----   |            | l         |           | p-op  |                        | buf addr+content&arr-args 7)                                                       |
| corr. MPI_REQUEST_FREE                                                                                                                                                                                       | ----f   | f          | l         |           | p-op  |                        |                                                                                    |
| MPI_COMM_CREATE                                                                                                                                                                                              | i-s-c-  | c          | nl        | b         | b-op  | C sq W1                | coll. over comm arg.                                                               |
| MPI_COMM_CREATE_GROUP                                                                                                                                                                                        | i-s-c-  | c          | nl        | b         | b-op  | C sq W1                | coll. over group arg.                                                              |
| MPI_COMM_DUP, MPI_COMM_DUP_WITH_INFO, MPI_COMM_SPLIT, MPI_COMM_SPLIT_TYPE,<br>MPI_CART_CREATE, MPI_GRAPH_CREATE, MPI_DIST_GRAPH_CREATE_ADJACENT, MPI_DIST_GRAPH_CREATE,<br>MPI_CART_SUB: see MPI_COMM_CREATE |         |            |           |           |       |                        |                                                                                    |
| MPI_INTERCOMM_CREATE,<br>MPI_INTERCOMM_MERGE                                                                                                                                                                 | i-s-c-  | c          | nl        | b         | b-op  | C sq W1                | coll. over union of<br>local & remote group                                        |
| MPI_COMM_IDUP                                                                                                                                                                                                | i-s---  | ic         | l         | nb        | nb-op | C sq                   | communicator handle                                                                |
| corresponding MPI_WAIT                                                                                                                                                                                       | ---c-   | c          | nl        |           | nb-op | C W2                   | 5)                                                                                 |
| corr. MPI_TEST returning flag=TRUE                                                                                                                                                                           | ---c-   | c          | l         |           | nb-op | C W2                   | 5)                                                                                 |
| corr. MPI_TEST returning flag=FALSE                                                                                                                                                                          | -----   |            | l         |           | nb-op |                        |                                                                                    |
| MPI_COMM_FREE, MPI_COMM_DISCONNECT                                                                                                                                                                           | ----f   | f          | nl        | b         | b-op  | C sq W1                | 19)                                                                                |
| MPI_INIT, MPI_INIT_THREAD                                                                                                                                                                                    | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                | collective over<br>MPI_COMM_WORLD                                                  |
| MPI_FINALIZE                                                                                                                                                                                                 | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                | collective over all connected<br>processes                                         |
| MPI_COMM_SPAWN, ..., _MULTIPLE                                                                                                                                                                               | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                | collective over<br>comm, 6)                                                        |
| MPI_COMM_ACCEPT, MPI_COMM_CONNECT                                                                                                                                                                            | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                | collective over comm                                                               |
| MPI_FILE_READ/WRITE[_AT SHARED],<br>MPI_FILE_DELETE/SEEK/GET_VIEW                                                                                                                                            | i-s-c-f | <b>c+f</b> | <b>l</b>  | <b>b</b>  | b-op  | -                      | <b>12)</b>                                                                         |
| MPI_FILE_READ/WRITE[_AT]_ <b>[ALL ORDERED]</b> ,<br>MPI_FILE_OPEN/CLOSE/SEEK_SHARED,<br>MPI_FILE_PREALLOCATE/SYNC,<br>MPI_FILE_SET_VIEW/SIZE/INFO/ATOMICITY                                                  | i-s-c-f | c+f        | nl        | b         | b-op  | C sq W1                |                                                                                    |
| MPI_FILE_IREAD/IWRITE[_AT SHARED]                                                                                                                                                                            | i-s---  | ic         | l         | nb        | nb-op | -                      | buffer 10a)                                                                        |
| MPI_FILE_IREAD/IWRITE[_AT]_ <b>ALL</b>                                                                                                                                                                       | i-s---  | ic         | l         | nb        | nb-op | C sq                   | buffer 10b)                                                                        |
| MPI_FILE_READ/WRITE[_AT]_ <b>ALL ORDERED_BEGIN</b>                                                                                                                                                           | i-s---  | <b>ic</b>  | <b>nl</b> | <b>b</b>  | b-op  | C sq W1                | buffer <b>8</b> )                                                                  |
| MPI_FILE_READ/WRITE[_AT]_ <b>ALL ORDERED_END</b>                                                                                                                                                             | ---c-f  | c+f        | nl        | b         | b-op  | C sq W1                |                                                                                    |